Cybersecurity Training Report
Instructor: Karan Bansod
Duration: 5 Days Training Program

Training Objective
To provide comprehensive knowledge of cybersecurity fundamentals, operating system security, network protection, protocol security, and email/browser security along with practical exposure to real-world implementations.

Day 1 – Module 1: Network Security Fundamentals
Topics Covered:
- Core Security Principles (Confidentiality, Integrity, Availability, Non-Repudiation)
- Risk, Threat, Vulnerability (RTV Model)
- Principle of Least Privilege (PoLP)
- Zero Trust Architecture
- Attack Surface (including IoT exposure)
- Types of Attacks:
  - Malware (Virus, Worm, Trojan, Spyware, Ransomware)
  - Social Engineering
  - MITM, DoS,SQL,XSS
- Backup & Restore (Full, Incremental, Differential)

Practical Coverage:
- Real-world attack scenario discussions (phishing, ransomware flow)
- Identification of malware types through case examples
- Backup strategy explanation and comparison
- Network Analysis with Wireshark.
- Attack Simulation using Burp Suite.

Day 2 – Module 2: OS Security & Access Control
Topics Covered:
- Client & Server Protection
- Separation of Services
- Hardening (OS, Application, Network)
- Patch Management
- Attack Surface Reduction
- Group Policy (GPO) concepts
- User Account Control (UAC)
- Secure Dynamic DNS (DDNS)
- Software Restriction Policies (SRP)

Authentication & Access Control:
- Multi-Factor Authentication (MFA)
- Password Policies
- Remote Access (RDP, VPN, SSH)
- Run As / Sudo
- User & Group Management (Local vs Domain)

Practical Coverage:
- Demonstration of GPO concepts and create Policies.
- Role-based access control examples
- Remote access methods demonstration
- User and group creation scenarios

- Linux commands & user creation and file management

Day 3 – Module 3: Network Device Secuirty

Topics Covered:

- Audit Policies & Log Management
- Types of auditing and log review
- Encryption Concepts:
    - Symmetric & Asymmetric
    - Public/Private Keys
- File Encryption (EFS)
- Drive Encryption (BitLocker)
- TPM & Digital Certificates

Network Defense:

- Firewalls (Stateful, Stateless, Proxy, NGFW)
- IDS vs IPS (NIDS, HIDS, NIPS, HIPS, WIPS)
- SIEM (conceptual)
- Content filtering, blacklisting/whitelisting
- Wireless Security & algorithms and Bands.

Additional Topics Covered:

- OSI Model (7 layers)
- TCP/IP Model
- IPv4 Addressing basics
- Ports & Protocol fundamentals
- Network sniffing concept

Practical Coverage:

- Log analysis examples
- Encryption behavior demonstration (file movement impact)
- Firewall rule logic explanation
- IDS vs IPS scenario-based demonstration
- WPA2 Enterprise Network Setup

Day 4 – Module 4: Network Device Security II

Topics Covered:

- Routing (Static & Dynamic)
- Routing Protocols (RIP, OSPF)
- Router Architecture (Control Plane & Data Plane)

Network Architecture Security:

- Honeypots & Honeynet
- DMZ (Perimeter Network)
- VPN (Remote Access & Site-to-Site)
- IPsec (AH, ESP, Tunnel Mode)
- Air-Gapped Networks
- DirectAccess

Protocol Security:

- DNSSEC
- SSH vs Telnet
- FTP, HTTP, HTTPS
- LDAP, SNMP

Practical Coverage:

- Cisco Packet Tracer:
    - VLAN configuration

    - NAT & PAT, SNMP, SSH setup
    - RIP routing configuration
- Basic network topology design and testing
- Protocol comparison (secure vs insecure)
- Protocol working concept with wireshark

Day 5 – Module 5: Email, Browser & Endpoint Security

Topics Covered:

- Email Protection:
    - Antispam
    - Spoofing, Phishing, Pharming
    - User Awareness & Training
- Email Security Mechanisms:
    - SPF, DKIM, DMARC
    - Email Header Analysis
- Browser Security:
    - Secure browser settings
    - Cache management
    - Private browsing
- Endpoint Security:
    - Antivirus & Anti-malware
    - Installation, updating, scanning
    - Remediation & alert investigation

Practical Coverage:

- Email header analysis demonstration
- Identification of phishing emails
- Antivirus scan configuration and scheduling
- Browser security settings review